Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.

q1 <- seq(1,55,5) ; q1# Generate a sequence of numbers, display the sequence
```

```
##  [1]  1  6 11 16 21 26 31 36 41 46 51
```

```
#2.

q1_mean <- mean(q1) ; q1_mean # Calculate the mean of the above sequence and assign it to a
```

```
## [1] 26
```

```
#variable named "question1_mean"
q1_median <- median(q1) ; q1_median # Calculate the median of the above sequence and assign it
```

```
## [1] 26
```

```r
#to a variable named "question1_median

#3.

q1_mean > q1_median # Determine if the mean is greater than the median using a logical operator
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```r
q5_i <- c("Jon", "Josh", "Janey", "Brooklyn") # String vector with student names
q5_ii <-  c(99, 85, 79, 92) # Numeric vector with student test scores
q5_iii <- c("TRUE", "TRUE", "FALSE", "FALSE") # Boolean vector with student scholarship status

# Create a data frame with student names, test scores, and scholarship status.
Student_Exam_Data <- data.frame(
  "Name" = q5_i,
  "Test Score" = q5_ii,
  "Scholarship Status" = q5_iii)
Student_Exam_Data
```

```
##       Name Test.Score Scholarship.Status
## 1      Jon         99               TRUE
## 2     Josh         85               TRUE
## 3    Janey         79              FALSE
## 4 Brooklyn         92              FALSE
```

9. QUESTION: How is this data frame different from a matrix?

   Answer: Data frames allow a user to create a table of data containing different data types in each column. A standard matrix requires that all data types be the same.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word "Pass"; otherwise print the word "Fail".
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead if `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```r
#10. Create a function using if...else

expanded_evaluation_10 <- function(x){ # Create an if-else statement to evaluate if
  #an input is greater than 50
  if (x > 50) {
    print ("Pass")}
  else {
    print ("Fail")}
}

#11. Create a function using ifelse()

ifelse_evaluation_11 <- function(x) { # Create and ifelse statement to evaluate
  #if an input is greater than 50
  ifelse(x > 50, "Pass", "Fail")
}

#12a. Run the first function with the value 52.5

expanded_evaluation_10(52.5) # Test the expanded if-else statement with 52.5
```

```
## [1] "Pass"
```

```r
#12b. Run the second function with the value 52.5

ifelse_evaluation_11(52.5) # Test the ifelse() function with 52.5
```

```
## [1] "Pass"
```

```r
#13a. Run the first function with the vector of test scores

test_vector <- seq(10,90,5) ; test_vector # Create a test sequence to be used in 13a and 13b
```

```
##  [1] 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90
```

```r
# expanded_evaluation_10(test_vector) # Run the function with the test vector

#13b. Run the second function with the vector of test scores
ifelse_evaluation_11(test_vector) # Run the function with the test vector
```

```
##  [1] "Fail" "Fail" "Fail" "Fail" "Fail" "Fail" "Fail" "Fail" "Fail" "Pass"
## [11] "Pass" "Pass" "Pass" "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for "R vectorization")

    Answer: The ifelse() function worked. This is because the function is designed to read through the vector (hence it is termed "vectorized"). A standard if-else statement is only designed to read a single value.

NOTE Before knitting, you'll need to comment out the call to the function in Q13 that does not work. (A document can't knit if the code it contains causes an error!)